1 Circular Orbit

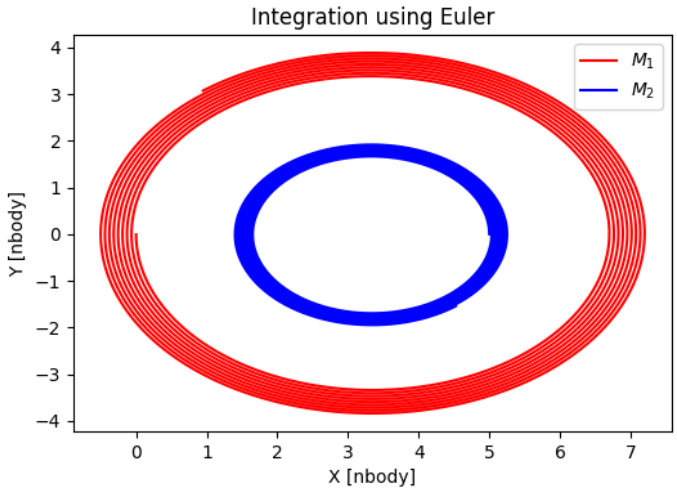


Figura 1: Integration of a circular orbit using the Euler method

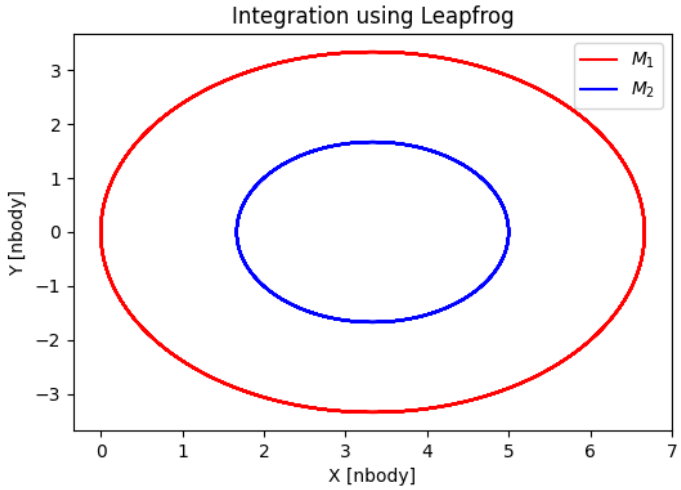


Figura 2: Integration of a circular orbit using the Leapfrog method

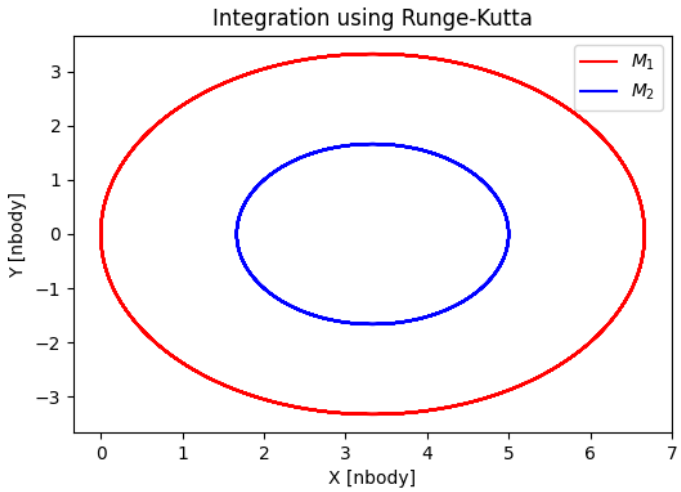


Figura 3: Integration of a circular orbit using the Runge-Kutta method

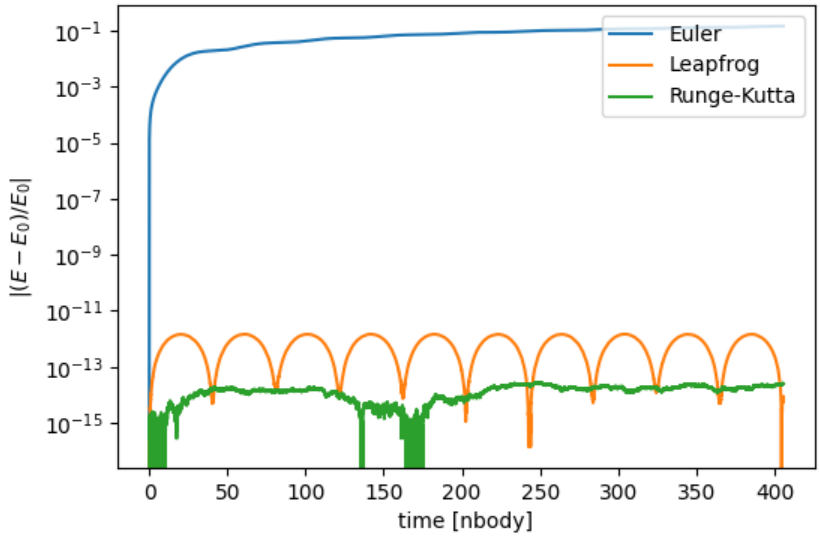


Figura 4: Energy error plot for each integration method

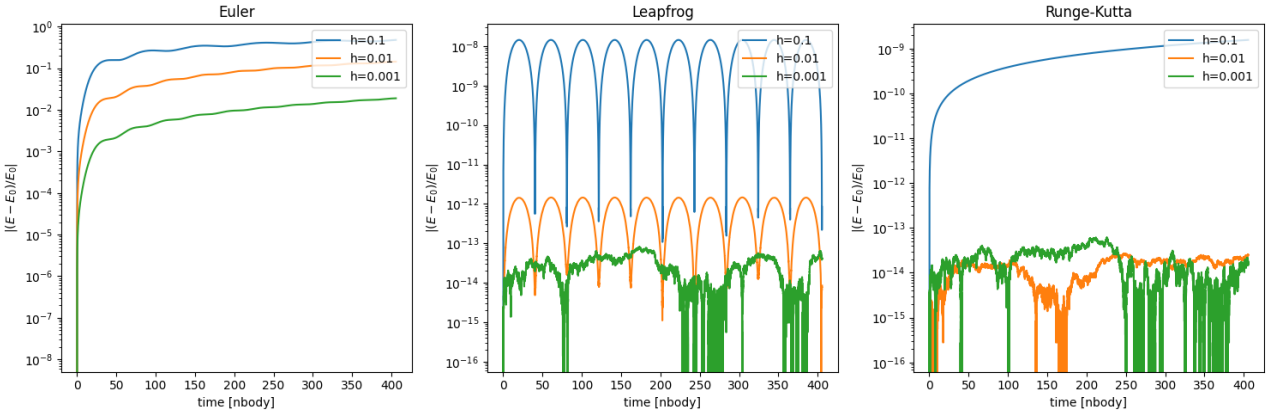


Figura 5: Energy error as function of the timestep

2 Eccentric Orbit

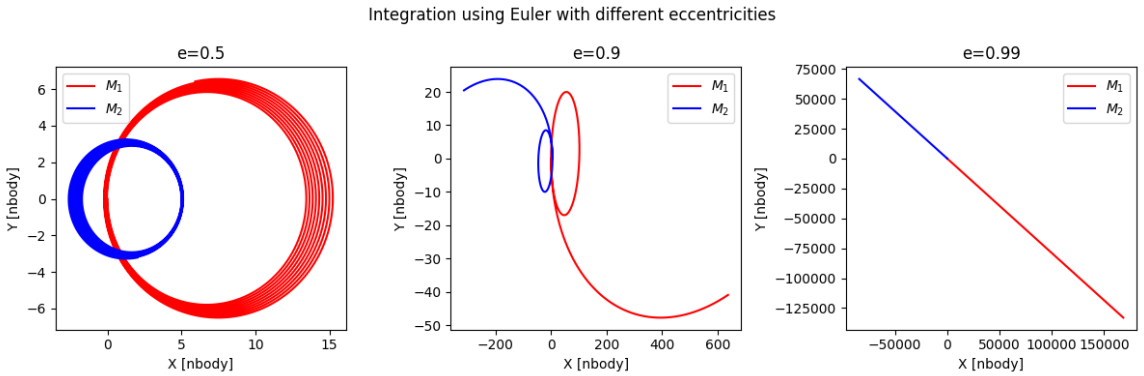


Figura 6: Integration of an eccentric orbit with three different values of eccentricity using the Euler method

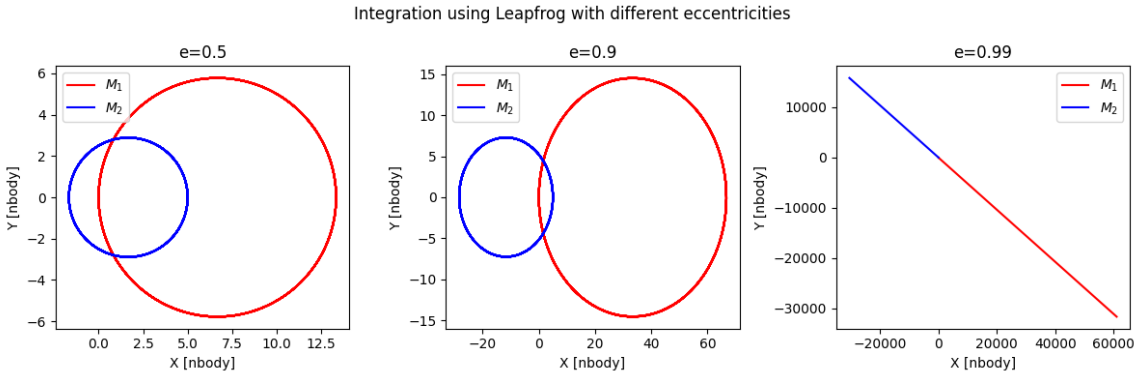


Figura 7: Integration of an eccentric orbit with three different values of eccentricity using the Leapfrog method

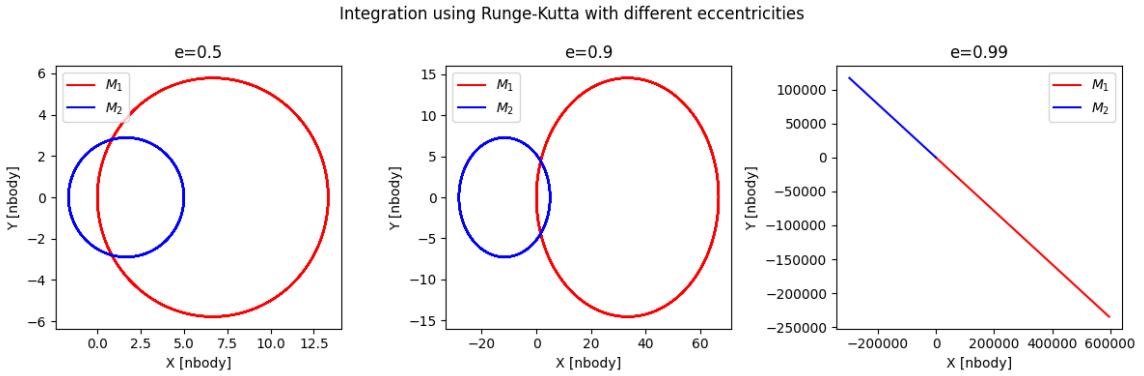


Figura 8: Integration of an eccentric orbit with three different values of eccentricity using the Runge-Kutta method

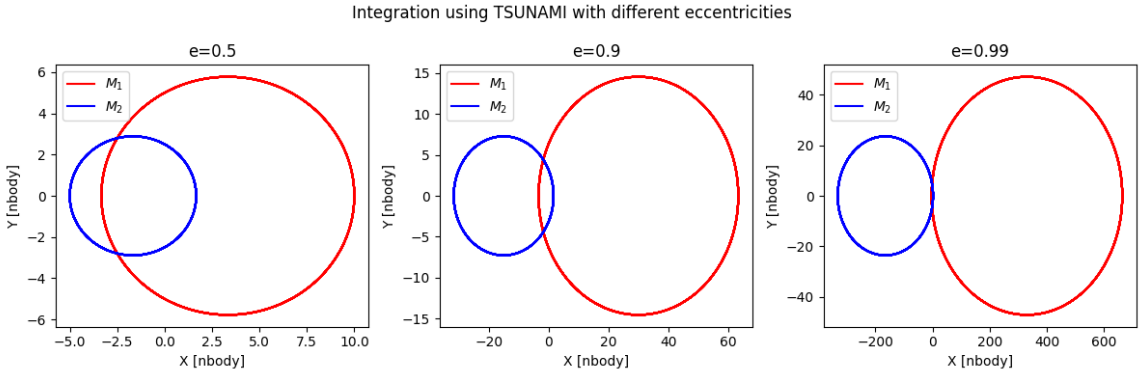


Figura 9: Integration of an eccentric orbit with three different values of eccentricity using the Tsunami integrator

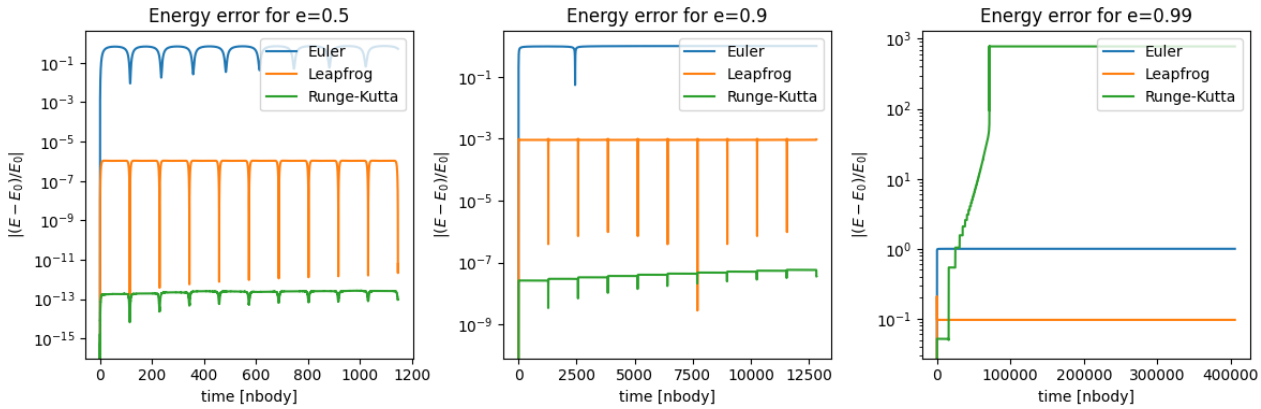


Figura 10: Energy error for every different eccentricity as function of the timestep

3 Adaptive integration

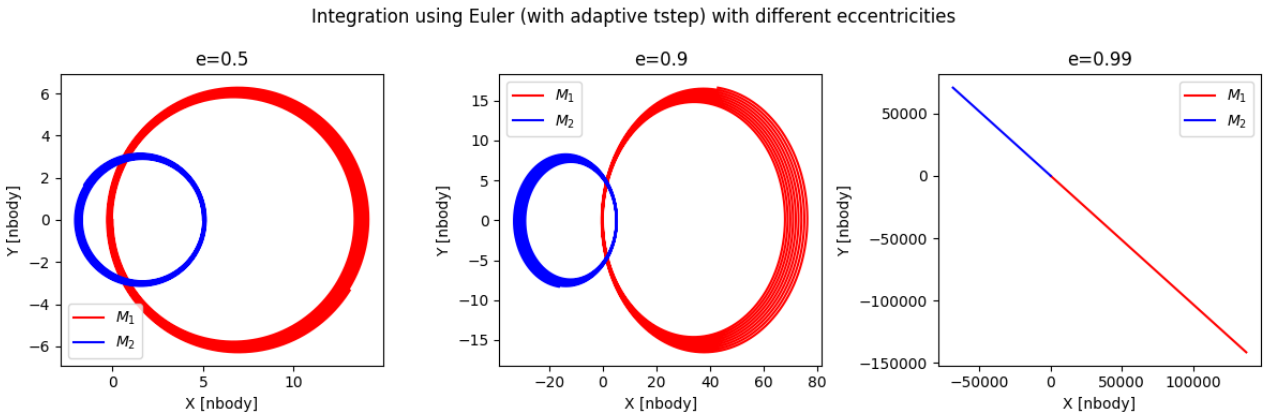


Figura 11: Integration of a circular orbit using the Euler method and the adaptive time step

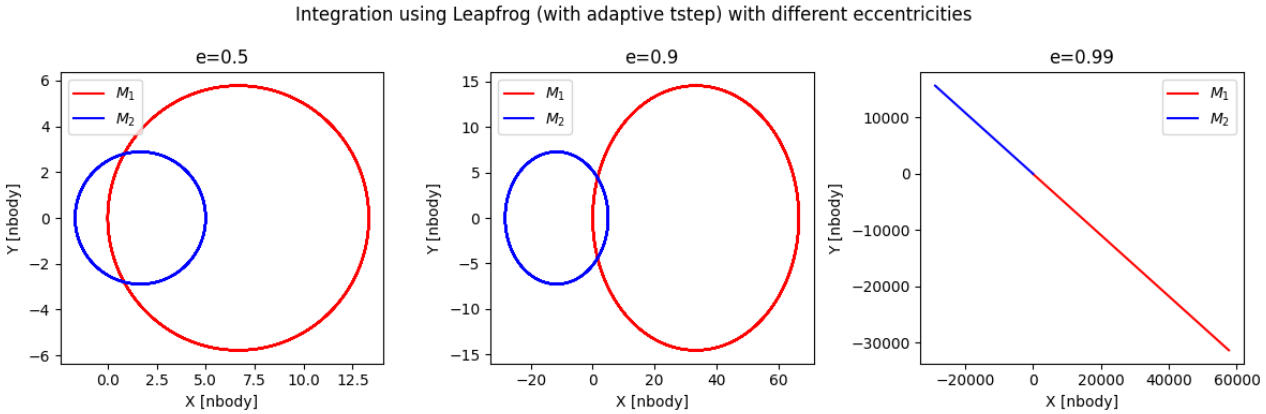


Figura 12: Integration of a circular orbit using the Leapfrog method and the adaptive time step

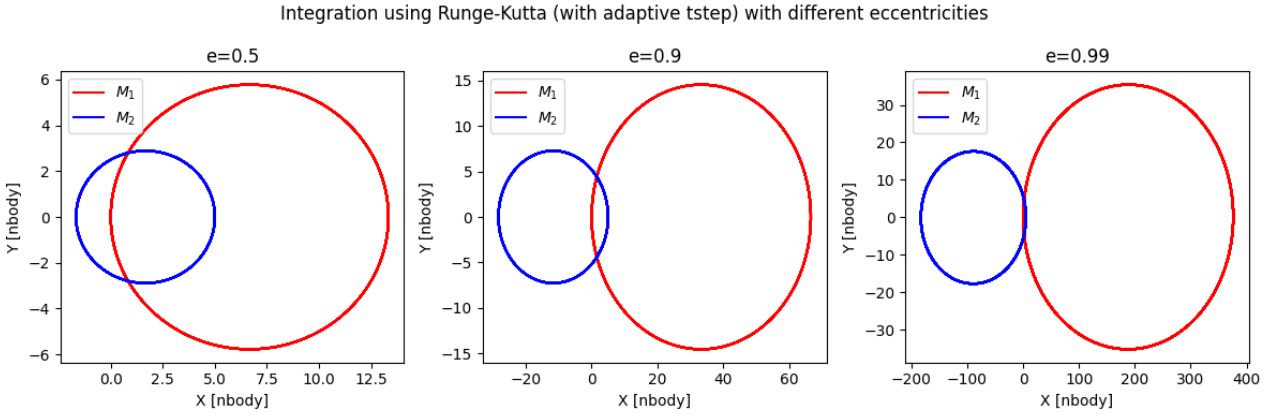


Figura 13: Integration of a circular orbit using the Runge-Kutta method and the adaptive time step

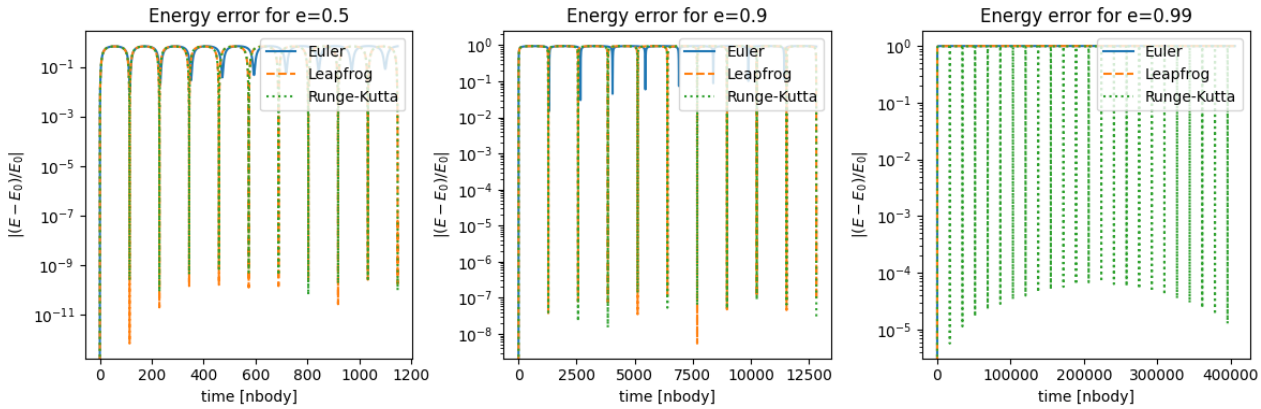


Figura 14: Energy error for every different eccentricity as function of the adaptive timestep

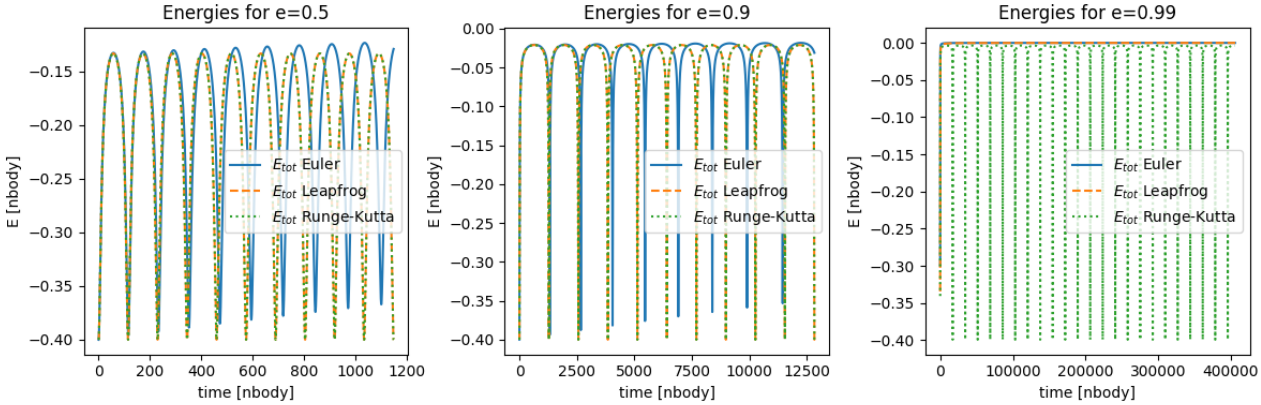


Figura 15: Total energy for every different eccentricity as function of the adaptive timestep

4 3-Body: Triangular Problem

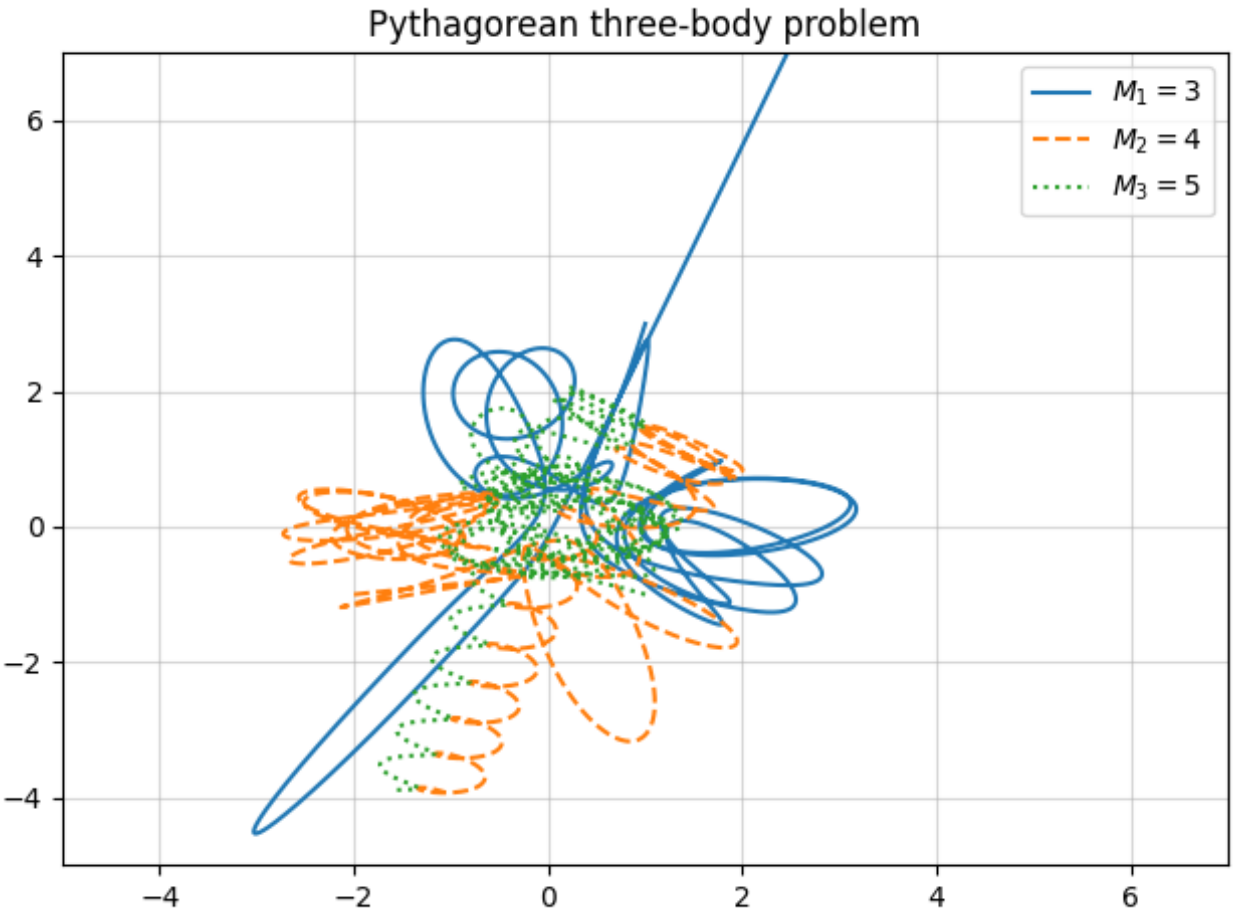


Figura 16: Integration of the 3-Body triangual problem using Tsunami integrator